HELIX Toolbox — Spall Detection & Qualification

This note describes the spall-detection and did-not-spall (DNS) qualification logic as implemented in the HELIX Toolbox, and explains every tuning parameter with a graphic example on a real 128 GS/s Ti PDV trace.

What the algorithm does

```
ALPSS  ─►  free-surface velocity (Savitzky–Golay smoothed) + uncertainty
  1. Uncertainty filter    drop points with rel. uncertainty ≥ 1.0
  2. Align t = 0           to ALPSS shock arrival (signal start)
  3. Window                keep spall_start … spall_end ns
  4. Gaussian smooth       one pass, spall_smoothing_sigma_ns
  5. Plateau               points ≥ plateau_threshold × peak → P1, P2, mean
  6. Pullback valley P3    first prominent minimum after the plateau …
  7. Recompression P4      … that is FOLLOWED BY a prominent peak (local)
  8. Gates                 4 recompression checks as a final refinement
     → first valley whose recompression passes = VALID SPALL, else DNS
```

The spall signature

A spall reflects a wave off the newly-formed internal fracture surface, so the measured velocity shows a two-sided fingerprint: a pullback dip (P3) followed by a recompression rebound (P4). Requiring BOTH — a prominent valley AND a prominent following peak — is the core of the qualification. A trace that simply shocks and releases (no rebound) is DNS.

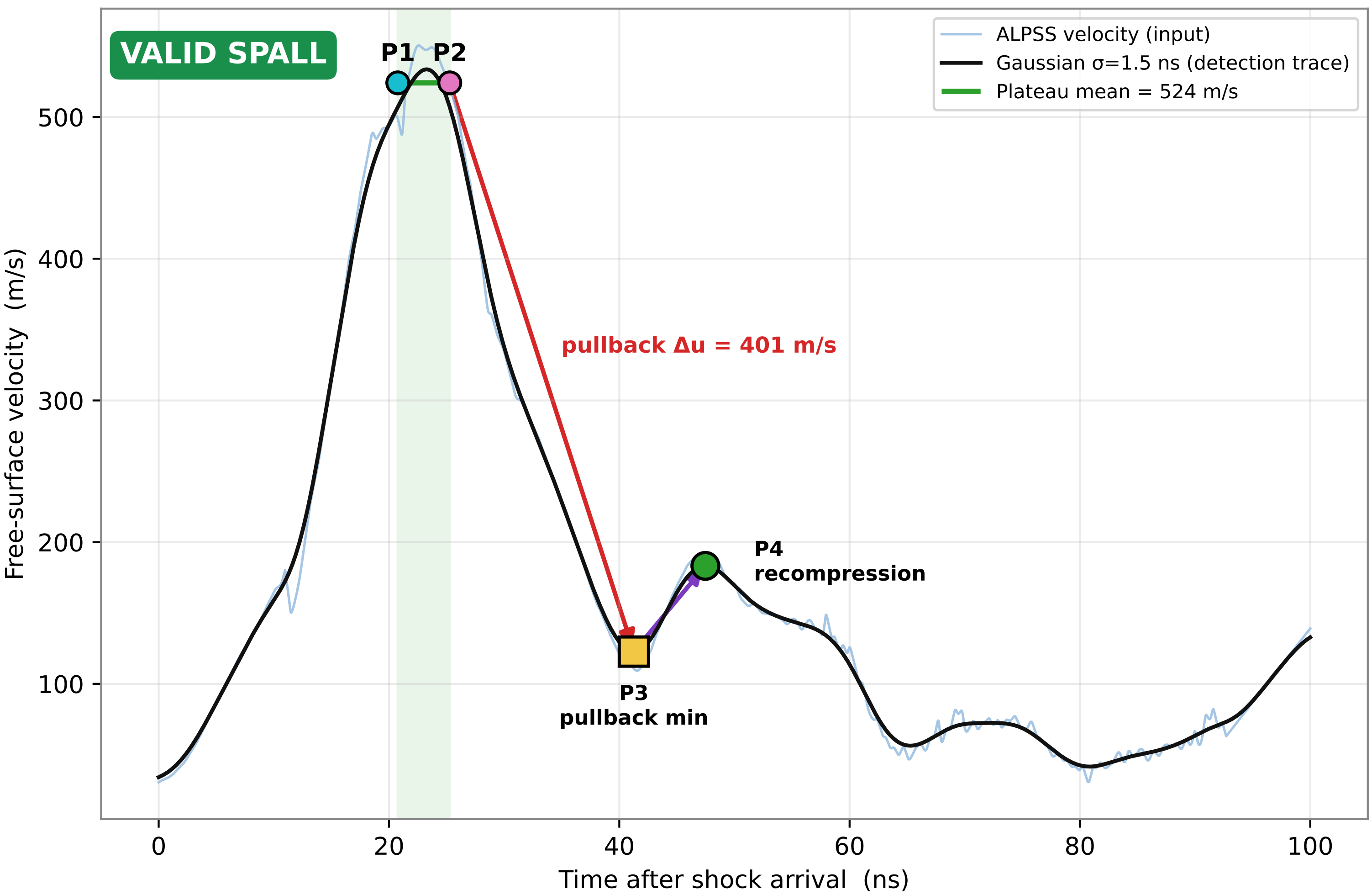
Physics (VALID spall traces only)

```
Spall strength   σ = ½·ρ·c_b·Δu_pullback          (Δu = plateau − P3)
Strain rate      ε̇ = |release slope| / (2·c_b)
Shock stress     σ = ρ·U_s·u_p ,  U_s = c + S·u_p ,  u_p = plateau/2
```

The following pages: (2) the algorithm on a real trace, (3) a parameter summary, then one graphic per tuning parameter.

The algorithm on a real trace (Ti)

JHAMAL00016-004_2026-04-23_21-52-09_shot20_ch1 — ρ=4507 kg/m³, c=4726 m/s, S=1.049



What the code computed here

```
Plateau  P1–P2   mean = 524 m/s                        →  shock stress
Pullback P3      123 m/s at 41.3 ns    Δu = 401 m/s    →  spall strength
Recomp.  P4      183 m/s at 47.5 ns    rebound = 60 m/s
```

All four recompression gates pass: P4>P3 (183>123); P4 ≥ P3×1.015 (183≥125); t(P4)−t(P3) ≥ 2.5 ns (6.2 ns); rebound ≥ 0.015×pullback (60 ≥ 6.0).

→ σ_spall = 4.275 GPa, ε̇ = 2.66e+06 s⁻¹, σ_shock = 5.907 GPa

Tuning parameters — summary

All live in the 'SPALL DETECTION FILTERS' block of the config

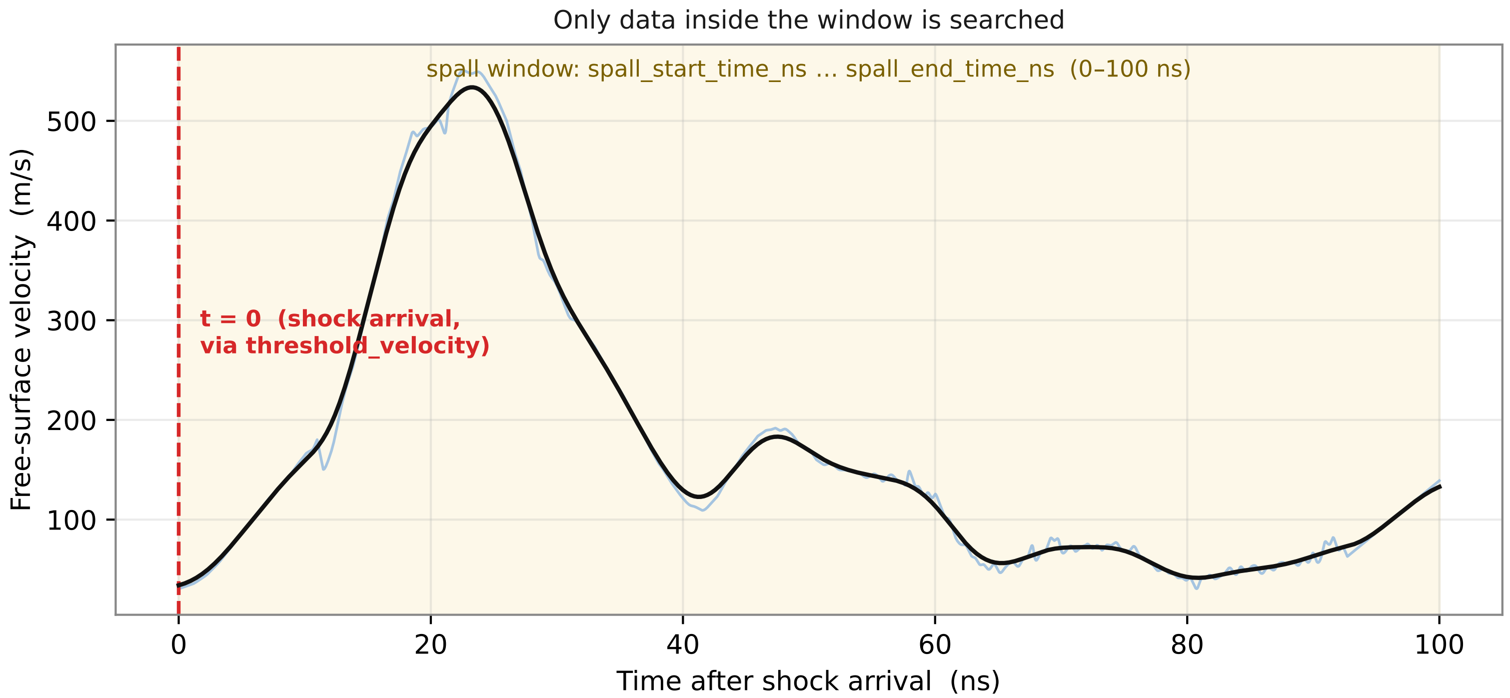
Parameter	Default	Controls
`plateau_threshold`	**0.95**	fraction of peak that counts as plateau
↑ *narrower plateau (near peak); ↓ wider, includes rise/fall*		
`spall_smoothing_sigma_ns`	**1.5**	Gaussian smoothing width (ns)
↑ *smoother but shaves peaks/fills valleys (σ_spall reads low)*		
`prominence_factor`	**0.01**	min valley/peak depth ÷ plateau
↑ *rejects weak features; ↓ catches shallow pullbacks + noise*		
`peak_distance_ns`	**3.0**	min time between detected features (ns)
↑ *merges nearby features; ↓ allows closely-spaced detections*		
`min_recomp_ratio`	**0.015**	rebound ÷ pullback depth (main gate)
↑ *stricter — needs a stronger recompression to accept*		
`min_recomp_velocity_ratio`	**1.015**	P4 ÷ P3 (rebound vs valley)
↑ *stricter on low-valley traces*		
`min_recomp_time_ns`	**2.5**	min rebound duration t(P4)−t(P3) (ns)
↑ *rejects fast spikes; ↓ allows quicker recompressions*		

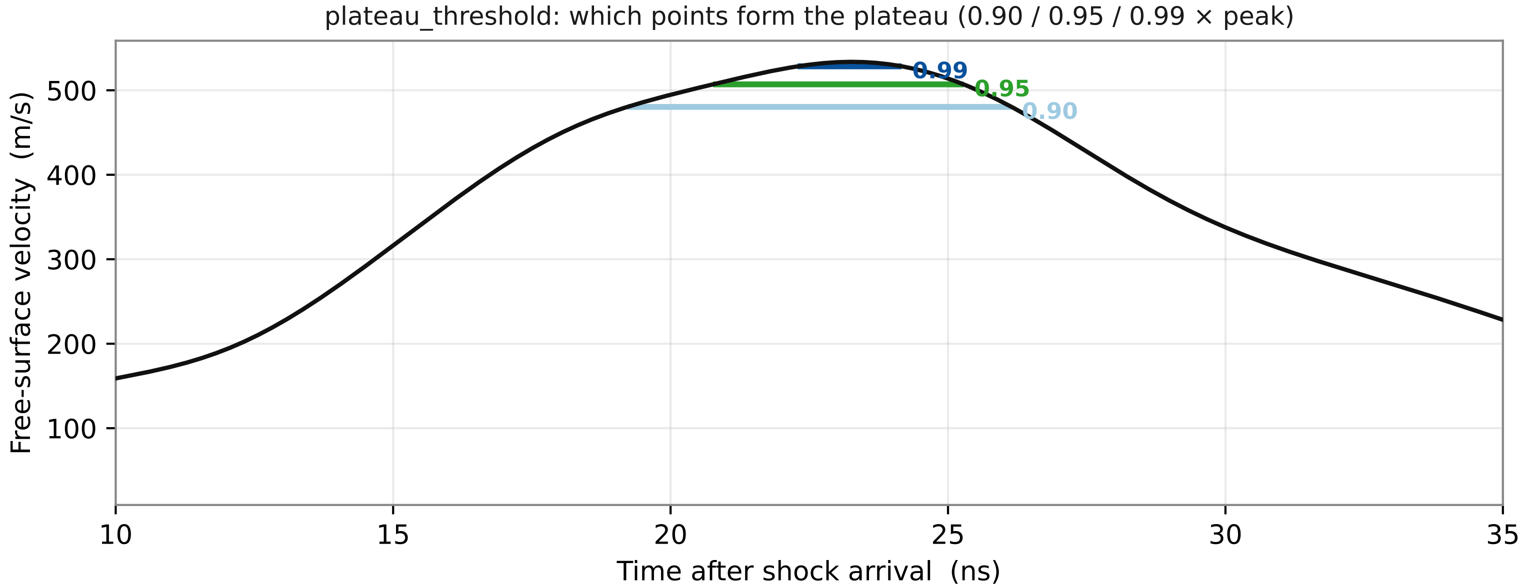
Analysis window

`spall_start / end_time_ns`	**0 / 100**	search window after t0 (ns)
`threshold_velocity_ms`	**5.0**	shock-arrival threshold for t0 (m/s)

Each parameter is illustrated on the following pages.

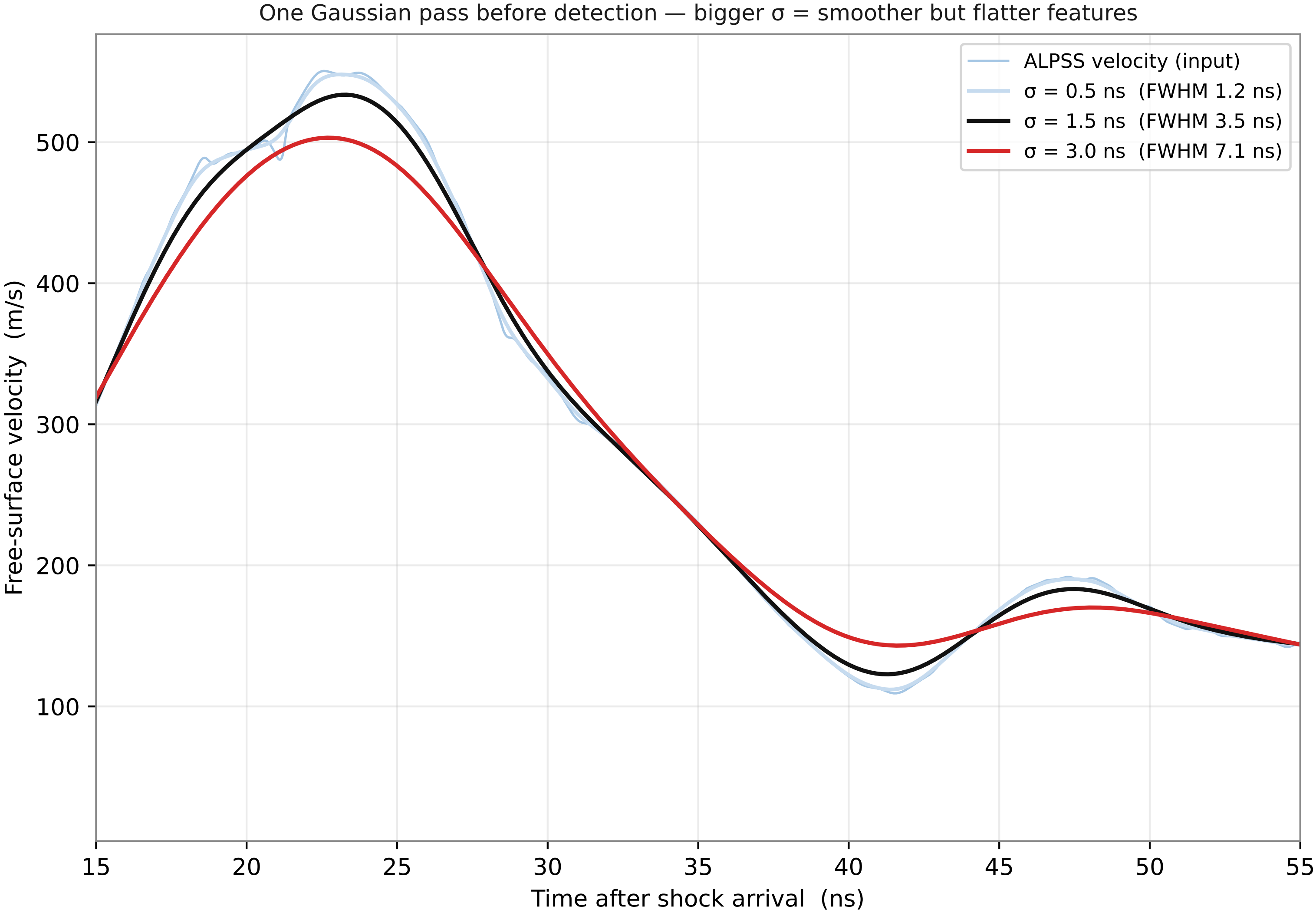
Parameter — analysis window & plateau_threshold





plateau_threshold sets the horizontal cut: every smoothed point at or above (threshold × peak) is a plateau point. Their mean is the plateau velocity that drives the shock stress and the pullback Δu (= plateau − P3). A higher value (0.99) hugs the peak — a narrow, noisier average; a lower value (0.90) widens the plateau into the rise and fall, biasing the mean downward. 0.95 is the default compromise. The window + threshold_velocity define WHERE to look and the t=0 origin; they rarely need changing unless shocks are weak or noisy.

Parameter — spall_smoothing_sigma_ns



The windowed trace is Gaussian-smoothed ONCE (this parameter) before P3/P4 detection; HEL detection is unaffected. σ is in nanoseconds of real time, so the physical smoothing scale is identical on any oscilloscope sample rate.

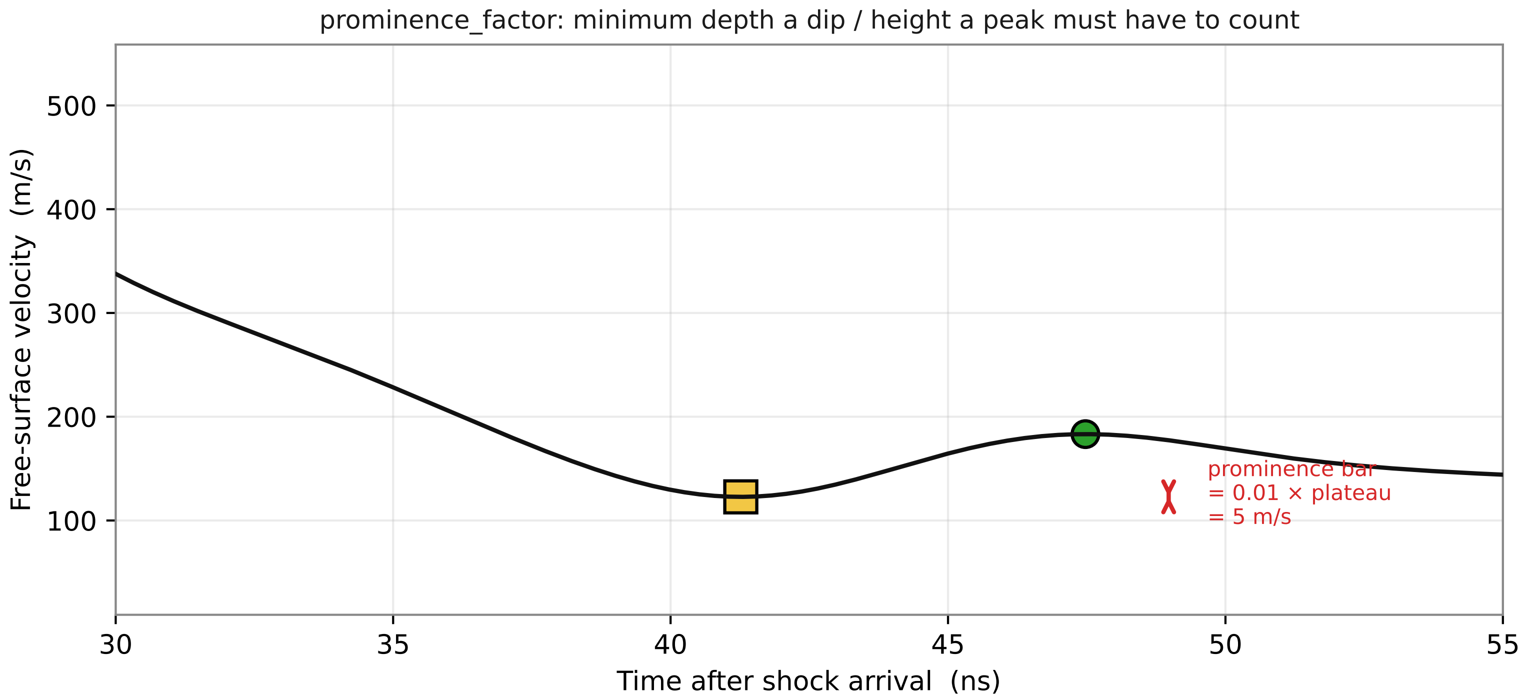
How to choose it

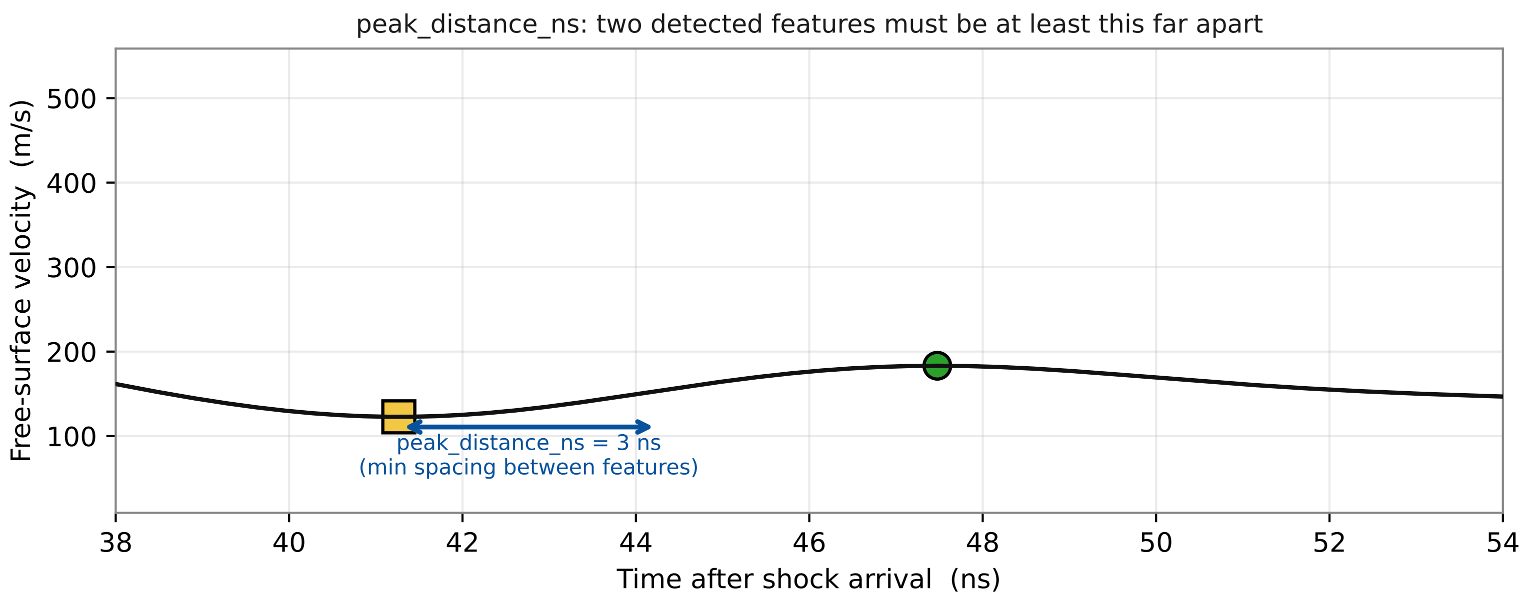
σ must sit between the noise scale (sub-ns here) and the narrowest real feature (this plateau is ~4–5 ns wide). FWHM ≈ 2.355 σ:

- σ = 0.5 ns → barely smooths; noise dips can survive and mislead P3.
- σ = 1.5 ns (default) → kills sub-ns noise, keeps peak/valley amplitude.
- σ = 3.0 ns → FWHM 7 ns is wider than the plateau: it shaves the peak and fills the valley, so the pullback (and σ_spall) reads systematically LOW.

Because spall values are read from this smoothed trace, over-smoothing directly lowers the reported spall strength — keep σ as small as the noise allows.

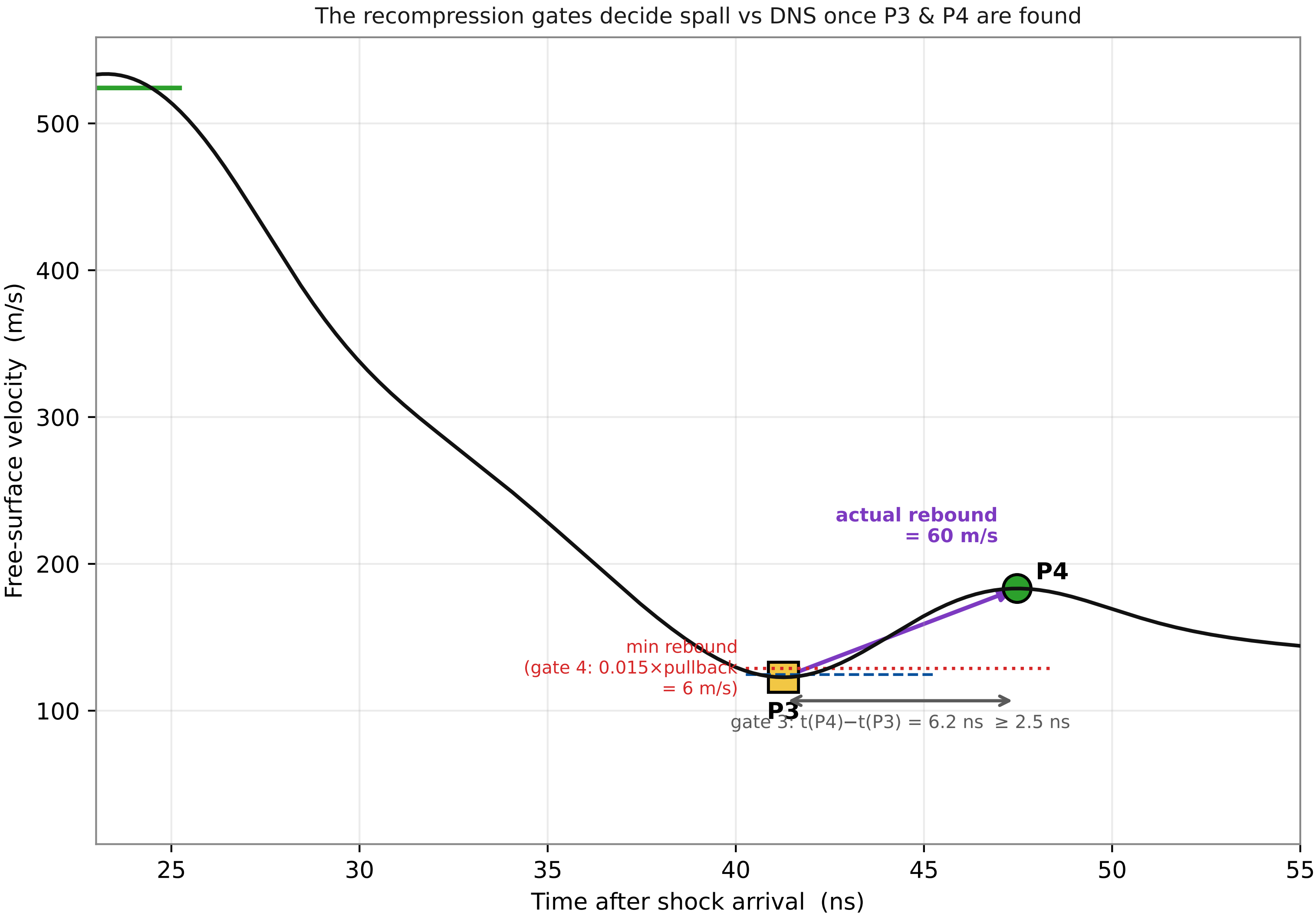
Parameters — prominence_factor & peak_distance_ns





prominence_factor is the DETECTION floor for both the pullback valley (P3) and the recompression peak (P4): a feature must stand out from its shoulders by at least prominence_factor × plateau (here 0.01 × 524 = 5 m/s). Lower it (0.005) to catch shallow/weak pullbacks; raise it (0.02) on ragged traces so a noise dimple is not detected. peak_distance_ns de-duplicates a single broad feature into one detection; keep it well below the ~6 ns pullback→recompression spacing so P3 and P4 are never merged.

Parameters — the three recompression gates



After P3 (valley) and P4 (recompression peak) are located, three thresholds confirm the rebound is a genuine spall recompression, not ringing:

```
min_recomp_ratio          0.015   rebound ≥ 1.5% of the pullback DEPTH (main)
min_recomp_velocity_ratio 1.015   P4 ≥ 1.5% above the VALLEY velocity
min_recomp_time_ns        2.5     P4 at least 2.5 ns after P3 (not a spike)
```

min_recomp_ratio is the primary physics gate — raise it to reject weak/ringing rebounds, lower it to keep marginal spall signals. min_recomp_velocity_ratio guards traces whose valley sits near baseline; min_recomp_time_ns rejects a fast noise spike right after the valley. Because the improved rule already requires a PROMINENT recompression peak for P3 to be accepted, these gates now act mostly as a final refinement — in practice a trace is usually resolved at the detection step (a prominent valley either has a prominent peak after it, or it does not).